

Technical Progress Sheet

Project: Smart Cold Chain Continuity System

Track B: Team – Selcius

Problem	Vaccines must stay within 2-8°C during transport, but many last-mile cold boxes have no built-in real-time monitoring at all - only passive insulation and ice-packs - so an excursion (e.g. the door being left open) can go unnoticed until the vaccines are already damaged. Even where monitoring devices are used, they only raise an alert after the fact, with no automated guidance on the best recovery action, especially where connectivity is too unreliable for a human coordinator to respond quickly. This results in avoidable vaccine wastage.
Proposed Solution	A low-cost IoT monitoring and response system, built as a digital twin in Wokwi. A DS18B20 sensor monitors box temperature, a push button reports whether the door is open or closed, and a NEO-6M GPS gives location. Data is buffered locally during network loss and sent over MQTT to a Node-RED dashboard. The system aims to support responses to a failure: rendezvous with another vehicle, return to origin, or redirect to a cold-storage facility.
Architecture	DS18B20 + door button + NEO-6M GPS -> ESP32 -> network link (simulated outages) -> MQTT broker -> Node-RED dashboard -> decision maker.
Current Progress	Wokwi circuit built with ESP32, DS18B20 and a door-status button, running our firmware. Working: temperature reading; door state detection; MQTT publishing / serial output; dashboard display. A simpler decision-support software that suggests a response, tested with simulated data scenarios .
Technology Stack	Platform: Wokwi (WiFi/MQTT simulation; LoRa not simulated, planned for post-simulation hardware phase). Hardware (simulated): ESP32, DS18B20, NEO-6M GPS, push button. Firmware: C/C++ (Arduino) with OneWire, DallasTemperature, TinyGPS++, PubSubClient. Software: MQTT broker (Mosquitto), Node-RED. Protocols: MQTT, 1-Wire, UART, GPIO.
Testing	Tested by varying three inputs and checking the output on a laptop screen: (1) Temperature — changed the simulated value and verified the correct reading, recommendation, and time to the nearest point were displayed. (2) Door — opened and closed the push button and verified the door status updated correctly. (3) GPS — changed the simulated location and verified the correct coordinates were reflected in the recommendation and time-to-nearest-point output. All values were viewed directly through the serial/console output on a laptop.
Limitations	Simulation does not capture real sensor noise, GPS conditions or power use. Temperature changes are simplified. The door is represented by a push button. Buffer capacity is limited. Decision mechanism not yet finalized.
Next Step	Develop a more advanced decision-support mechanism and test it with real transport data, door status, location and situation, then run more parameter tests and collect graphs and logs to validate results before the final.

Summary: The project delivers a simulated, low-cost IoT-based monitoring and response system that detects cold-chain problems and provides the information needed for timely decisions, aiming to reduce vaccine wastage during transport.